

Decomposing Advice Markers Across OLMo-2 Checkpoints and Training Data

Anonymous ACL submission

Abstract

Instruction-tuned models often combine empathy, validation, professional referral, and structuring when giving personal advice. We trace these four markers across OLMo-2’s public Base, SFT, DPO, and RLVR checkpoints and exact training data. We evaluate 140 prompts in five conditions with five final-model generations per prompt and add four-generation matched controls at Base, SFT, and DPO. Final-model controls decompose the apparent template: empathy tracks emotion, structuring tracks how-to form, referral tracks advice and health, and only validation is elevated on advice against every control. Six of eight advice-minus-factual gap increases at SFT survive correction. The stricter matched-stage analysis leaves four of 24 interactions significant: 7B SFT makes empathy more advice-specific, DPO reverses this in both sizes, and 7B SFT shifts structuring toward neutral advice. Only the two 7B empathy effects persist across all detector thresholds. Exact searches find referral and structuring phrases enriched in DPO chosen-only pairs. A separate 120-sentence AI consensus audit finds macro precision 0.28 and recall 0.72, so checkpoint rates remain provisional and paired phrase enrichment is the strongest evidence.

1 Introduction

Ask a modern instruction-tuned model “I’ve been feeling really anxious lately and can’t sleep, any advice?” and you will, in our sample, reliably receive a structured response: an acknowledgment, validation of the user’s emotional state, suggestion to seek professional help, and list-organizing language. Two questions follow. First, is this a single *advice template*, or several behaviors that merely co-occur on advice prompts? Second, *where* in training does each behavior emerge?

Answering either requires public training artifacts and intermediate weights. A final OLMo-2 Instruct checkpoint is the product of four data-bearing

stages: pretraining, supervised fine-tuning (SFT), direct preference optimization (DPO), and reinforcement learning with verifiable rewards (RLVR) (Team OLMo et al., 2025). OLMo-2 offers an unusually complete public chain of post-training data and intermediate weight checkpoints. We exploit both: we run the same behavioral evaluation on every public stage checkpoint, and we search the available indexed data: the Stage-1 corpus OLMo-Mix-1124 via infini-gram, the model-specific Tulu-3 SFT mixtures, DPO preference mixes, and RLVR sets.

We are deliberate about what this evidence is: **observational co-occurrence and stage localization, not causal attribution**. We measure where behaviors appear across checkpoints and where their phrasings live in the corresponding data; we do not retrain. Our data search is not fully end-to-end: Stage-2 mid-training (Dolmino-Mix-1124) has no public index and is not searched.

Contributions.

1. **An auditable behavior-level tracer on exact open data.** A 140-prompt battery in five matched conditions (advice, factual, emotional non-advice, neutral advice, domain factual), five final-model generations per prompt, a released lexicon+e5 detector, role-scoped corpus search, and a separate 120-sentence AI consensus audit.
2. **A controlled decomposition of the “advice template.”** At the primary threshold, detector-positive empathy tracks emotional content, structuring language tracks the how-to form, referral markers track advice/health, and the validation marker alone is elevated against every matched control; the threshold sweep identifies which distinctions are detector-sensitive (Section 4.1).
3. **Stagewise difference-in-differences with**

matched specificity tests. Advice-minus-factual trajectories show six of eight SFT gap increases survive correction; DPO widens validation in both sizes and referral at 1B while collapsing the structuring gap. A new 24-test Base/SFT/DPO matched-control extension identifies which changes are specific to emotion, advice form, or health domain (Section 4.2).

4. **A measurement audit rather than an assumed oracle.** Two independent blinded AI passes agree exactly on 117/120 sentences; Claude Opus 4.8 adjudicates the remaining three. The resulting low precision sharply limits output-level interpretation, while the paired DPO and exact corpus results remain independent of the embedding threshold at the phrase level.
5. **Compatible data-layer evidence with paired inference.** Referral phrasings cluster in exact-SFT assistant turns (phrase-sparse, leave-one-out reported) and are more frequent in DPO chosen-only than rejected-only pairs under McNemar tests; descriptive per-word ratios agree. Searched RLVR data have near-zero expressive and referral counts but retain generic structuring markers. Verbatim-only novelty is stated as a first-class limitation (Sections 4.3–4.4).

2 Related Work

Behavioral tests and checkpoint trajectories. Our matched prompt battery follows the behavioral-testing principle that targeted capabilities require controlled tests beyond aggregate accuracy (Ribeiro et al., 2020). Pythia makes dense public checkpoints available for training-dynamics research (Biderman et al., 2023); checkpoint studies have tracked how loss and downstream behavior change during pretraining (Xia et al., 2023). We instead compare named post-training stages and test changes in condition gaps rather than raw behavior rates.

Pretraining attribution and n-gram search. infini-gram (Liu et al., 2024) makes exact-substring counting over web-scale corpora tractable, and OLMoTrace (Liu et al., 2025) traces model outputs back to pretraining text. Both are pretraining-only; we add role-scoped search over the post-training layers, where our stagewise results locate

the change.

Open instruction and preference data. Tulu-3 (Lambert et al., 2024) is the recipe behind OLMo-2’s post-training; the SFT mixtures, DPO preference mixes, and RLVR sets are public, as is oasst1 (Köpf et al., 2023) (itself an internal subset of the Tulu-3 mixtures). Public data at every stage is what makes the trace auditable.

The superficial alignment hypothesis. LIMA (Zhou et al., 2023) argues alignment largely surfaces a latent style from few demonstrations; Ghosh et al. (2024) interrogate this. Our stagewise gaps give a behavior-level version: each marker is nonzero at base, and post-training produces larger advice-minus-factual detector gaps.

Causal data attribution. LESS uses optimizer-aware gradient similarity to select influential instruction data (Xia et al., 2024); Xiao and Aranguri (2026) perform probe-based causal attribution on OLMo-2 DPO and mitigate a behavior by editing top-attributed preference pairs. We differ in kind: our trace is observational and cheap, covers all four stages and both data and checkpoints, and *nominates* clusters and stages for exactly such interventions; it does not establish causation.

3 Method

3.1 Behavior detection (lexicon + e5)

A hand-seeded, inspectable lexicon defines four reported constructs: **empathy expression** (artifact key `empathy_opener`), **validation**, **professional referral** (`disclaimer`), and **structuring language** (`structure`). A fifth category, `crisis_referral`, fails the 120-sentence audit and is excluded. Exact matching is augmented with **e5-small** paraphrase embeddings (Wang et al., 2024), a response is detector-positive when any sentence exceeds similarity $\tau = 0.82$ to a seed exemplar (sweep 0.80/0.82/0.84). **Detector construction and circularity.** The lexicon, exemplars, and threshold were hand-seeded during an 8-prompt pilot on OLMo-2-1B and frozen *before* the present 140-prompt battery was authored; all three specialized matched controls (emotional non-advice, neutral advice, and domain factual) and 100 of the 140 prompts were written after freezing. The sample sizes refer to distinct datasets: the 140 prompts form the behavioral evaluation, whereas 120 stratified final-model sentences form a detector-accuracy

audit. We audit those sentences (40/30/20/20/10 across the five conditions) using two complete, independent, blinded AI passes and Claude Opus 4.8 adjudication of their three disagreements. This is an AI diagnostic, not human validation. The labels, adjudication reasons, and detector are released (Section 4.4; Appendix B).

3.2 Prompt battery and generation

Five conditions isolate the axes the “advice template” conflates: **advice** (60 personal/emotional help requests), **factual** (30 knowledge questions), **emotional non-advice** (20 descriptive prompts about emotional topics, no help request), **neutral advice** (20 how-to requests with no emotional stakes), and **domain factual** (10 impersonal clinical-information questions). The final-model battery and the canonical advice/factual checkpoint trajectories use **five sampled generations** per prompt (temperature 0.7, top- p 0.9, 256 new tokens, distinct seeds). The reviewer-driven Base/SFT/DPO matched-control extension uses four; its balanced comparisons use seeds 0–3 for every condition.

3.3 Statistical unit

The clustering unit is the **prompt**: each prompt contributes its mean detector-positive indicator over its four or five generations, and condition rates are means over prompts. All between-condition inference uses **prompt-clustered permutation tests** (10,000 permutations of prompt-level means; two-sided; Monte Carlo $p = (b + 1)/(B + 1)$). Benjamini–Hochberg correction is applied separately to the 24 final-condition contrasts, 28 advice/factual adjacent-stage interactions, 24 matched-control stage interactions, and 8 DPO pairwise tests. We do *not* treat generations as independent Bernoulli trials: Fisher/Wilson statistics computed on “equivalent counts” are reported only as an appendix sensitivity analysis with that caveat (Appendix E).

3.4 Stagewise evaluation and stage-data search

The same evaluation runs on every public checkpoint: 1B Base→SFT→DPO→RLVR1→Instruct; 7B Base→SFT→DPO→Instruct. Base checkpoints have no chat template and use a plain Question/Answer format (a comparison caveat we state). The stagewise grid runs the 90 advice+factual prompts at every checkpoint.

All five conditions run on the final models and, in the four-seed specificity extension, at Base, SFT, and DPO. For each stage’s available indexed data we search: the exact Stage-1 corpus (v4_olmo-mix-1124_llama; novelty = 1–verbatim coverage), each model’s exact SFT mixture (tulu-3-sft-olmo-2-mixture-0225 for 1B; tulu-3-sft-olmo-2-mixture for 7B) in DuckDB with **assistant-only** scoping (866,138 / 939,344 conversations for 1B/7B; oasst1 not searched separately since it is an internal subset), the DPO preference mixes (pair-level chosen-only/rejected-only counts, exact McNemar tests, and phrase occurrences per million response words), and the RLVR sets. SFT recoverability compares behavioral phrases against **length-matched topical controls** from the same answers, per-million documents, with a distinct-phrase leave-one-out.

Reproducibility. Stagewise, condition, and corpus numbers derive from `summary_v3.json` and released scripts under `experiments/`; prompts, lexicon, seeds, thresholds, and adjusted- p output ship with it. Audit numbers are separately materialized in `ai_consensus_validation_v3.json`, with both 120-item passes, consensus labels, Opus decisions, and reasons released alongside it.

4 Results

Results at a glance. Four outcomes organize the analysis. First, the apparent advice bundle decomposes into affect-linked empathy, form-linked structuring, domain-linked referral, and a validation marker consistently elevated on advice. Second, all eight advice-minus-factual detector gaps widen at SFT (six survive correction), while matched-stage controls show that only a subset of these changes is advice-specific. DPO widens validation gaps but spreads structuring to factual answers and empathy toward emotional non-advice. Third, exact phrase searches find referral and structuring enrichment in DPO chosen-only pairs; searched RLVR data have near-zero expressive and referral counts but retain generic structuring markers. Fourth, the separate 120-sentence audit finds low detector precision, so the checkpoint results are marker trajectories rather than estimates of semantic behavior prevalence.

4.1 The detector-positive advice bundle decomposes at the primary threshold

Table 1 gives mean prompt-level detector-positive rates per condition on the final models; Table 2 tests advice against each matched control. Against *plain factual* controls every detector category separates sharply in both models (all prompt-clustered $p \leq .0002$; referral 0.900 vs 0.080 at 1B, 0.913 vs 0.060 at 7B). But the matched controls show this is **not one advice-specific template**:

- **Empathy expression tracks emotional content at $\tau = .82$.** It fires *more* on emotional non-advice descriptions than on advice (0.79 vs 0.590 at 1B, $p = .010$; 0.76 vs 0.617 at 7B, $p = .098$), and near-zero on neutral advice (0.02–0.07).
- **Structuring language tracks the how-to form.** It is at ceiling on *neutral* advice exactly as on personal advice (diff 0.00, $p = 1.0$ at 1B; -0.03 , $p = .57$ at 7B).
- **Professional referral tracks the advice/health domain at $\tau = .82$.** It fires on 0.72–0.74 of neutral-advice and 0.82 of clinical-information prompts; its advice increment over domain-factual is small and not significant ($+0.080$, $p = .352$ at 1B; $+0.093$, $p = .159$ at 7B).
- **The validation marker is the only detector category consistently elevated on personal advice relative to every matched control:** higher on advice than *every* control in *both* models (all unadjusted $p \leq .0044$; all BH-adjusted $q \leq .0082$), including emotional non-advice (0.363 vs 0.120, $p = .0027$; 0.383 vs 0.13, $p = .003$).

Within the frozen detector, the primary-threshold marker bundle separates into affect-linked empathy, form-linked structuring, domain-linked referral, and a validation marker consistently elevated on advice. These are not detector-invariant universals: the threshold sweep preserves the structuring result but qualifies the empathy and referral contrasts (Section 4.4). The 1B-vs-7B differences are small and mixed; we make no scaling claim.

4.2 Stagewise marker dynamics and matched specificity

Raw advice-side rates rising across stages would be ambiguous because post-training could simply

make *all* outputs more cautious and listy. The right quantity is the **advice-minus-factual gap** Δ_s at each stage (Table 4) and its *change* across adjacent stages, a stage $\times$ condition difference-in-differences tested with a prompt-clustered permutation test on per-prompt stage trajectories (Table 3; 10k permutations, $n=60$ advice/30 factual). Three findings.

- **All SFT gap estimates are positive.** Six of eight base $\rightarrow$ SFT changes survive BH correction (referral $+0.25/+0.24$, structuring $+0.34/+0.23$); validation does not ($q = .137/.121$). Every detector category is already nonzero at base, so SFT widens advice-minus-factual marker gaps for categories already detected at Base.
- **DPO acts marker-specifically.** It significantly further widens validation ($+0.21/+0.15$, $q = .002/.021$) and 1B referral ($+0.19$, $q = .002$), while 1B empathy is only a tendency ($+0.12$, $q = .121$) and 7B empathy/referral do not change significantly. Its preference data nonetheless contains more referral phrasings in chosen than rejected responses (Table 5), consistent with *maintaining* that near-saturated gap.
- **DPO collapses the structuring gap.** Its gap falls $-0.37/-0.28$ at DPO ($p = .0002$), not because advice-side structuring drops (it is at ceiling) but because DPO raises structuring on *factual* answers from 0.07 to 0.54–0.59. Preference tuning spreads structuring language everywhere while leaving the other advice-minus-factual detector gaps positive.

RLVR and the final merge move no gap significantly (every dpo $\rightarrow$ rlvr and final-merge transition has $p \geq 0.168$), consistent with RLVR’s math-only data (Section 4.3). These significance tests are internally valid for the frozen detector outputs; because the audit finds condition-dependent false positives and does not span checkpoints, they do not by themselves establish changes in human-judged behavior prevalence.

Matched controls qualify advice specificity. The four-seed Base/SFT/DPO extension tests 24 planned stage $\times$ control interactions (Appendix D). Four survive BH correction at $\tau = .82$. For empathy relative to emotional non-advice, 7B SFT increases the advice gap ($+0.358$, $q = .016$), but DPO reverses it at both 1B (-0.467 , $q = .001$)

Behavior	Model	Advice	Factual	Emotional	Neutral adv.	Domain fact.
empathy expr.	1B	0.590	0.040	0.790	0.020	0.340
	7B	0.617	0.027	0.760	0.070	0.360
validation	1B	0.363	0.007	0.120	0.000	0.040
	7B	0.383	0.013	0.130	0.030	0.060
referral	1B	0.900	0.080	0.510	0.740	0.820
	7B	0.913	0.060	0.540	0.720	0.820
structuring	1B	1.000	0.580	0.820	1.000	0.920
	7B	0.970	0.547	0.830	1.000	0.840

Table 1: Mean prompt-level detector-positive rate per condition (final models, $\tau = 0.82$; each prompt averaged over 5 generations). Bold marks the cells that break the “advice-specific” reading: empathy peaks on emotional *non-advice* prompts, and structuring language is at ceiling on *neutral* advice. The audit sample shows substantial overprediction, suggesting upward-biased absolute rates (Section 4.4); *crisis_referral* is excluded (0 TP / 11 FP in the 120-sentence audit).

Behavior	M	−Emot.	−Neut.adv.	−Dom.fact.
empathy	1B	−.200 (.010)	.570 (.0002)	.250 (.016)
	7B	−.143 (.098)	.547 (.0002)	.257 (.031)
validation	1B	.243 (.003)	.363 (.0002)	.323 (.003)
	7B	.253 (.003)	.353 (.0003)	.323 (.004)
referral	1B	.390 (.0002)	.160 (.006)	.080 (.352)
	7B	.373 (.0002)	.193 (.001)	.093 (.159)
structuring	1B	.180 (.0002)	.000 (1.0)	.080 (.018)
	7B	.140 (.004)	−.030 (.574)	.130 (.026)

Table 2: Advice minus each matched control (prompt-clustered permutation p , 10k draws). Benjamini–Hochberg correction across the 24 displayed tests leaves the validation conclusion unchanged (all $q \leq .0082$); non-significant and negative cells carry the decomposition (Section 4.1).

Model	Behavior	base→SFT	SFT→DPO
1B	empathy	+0.190 (.002)	+0.117 (.055)
1B	validation	+0.093 (.068)	+0.213 (.0003)
1B	referral	+0.253 (.0004)	+0.190 (.0006)
1B	structuring	+0.337 (.0002)	−0.367 (.0002)
7B	empathy	+0.263 (.0007)	+0.070 (.251)
7B	validation	+0.103 (.056)	+0.147 (.008)
7B	referral	+0.243 (.0003)	+0.083 (.109)
7B	structuring	+0.227 (.0005)	−0.277 (.0002)

Table 3: Change in the advice-minus-factual gap across transitions (plus-one-corrected prompt-clustered permutation p , 10k draws). BH correction across all 28 adjacent-stage×behavior tests preserves six of eight SFT effects (validation: $q = .137/.121$), both DPO validation effects, 1B referral, and both structuring collapses; later raw $p \geq .168$.

and 7B (−0.642, $q = .001$). At 7B, SFT also shifts structuring toward neutral advice (−0.208, $q = .026$). No validation or referral interaction survives $q < .05$; 1B DPO validation tendencies have $q = .059$ –.095. The threshold sweep yields 3/4/5 significant interactions at .80/.82/.84; only the two 7B empathy transitions survive at all three thresholds. Thus the broad advice-minus-factual changes cannot all be read as advice-specific.

4.3 The data layers provide compatible phrase-level evidence

SFT: a finite, auditable phrase-matching set (phrase-sparse). Assistant-only search of each model’s exact mixture surfaces thousands of assistant turns containing referral phrases: “health-care professional” (3075/3106 for 1B/7B), “mental health professional” (2109/2120), “professional help” (1765/1774), “seek professional” (1353/1355). Length-matched recoverability is above topical controls at ≥ 3 words (3-word: $20.5\times$ at 1B, $7.3\times$ at 7B; 4-word: $136.9\times/14.5\times$)

but **phrase-sparse**: only 3/6 distinct 3-word types, dominated by the list preamble “here are some.” The distinct-phrase leave-one-out is the honest figure: dropping that one phrase lowers the 3-word mean from 5455.2 to **1305.2/M** (1B) and 3026.0 to **1065.9/M** (7B) (Appendix F). Any inference must use the 3–6 distinct types as its unit, not the 45/55 occurrences. Role-scoping remains necessary: blob search inflates “talk to a” 1.41 – $1.43\times$ via user turns.

DPO: paired inference and descriptive length normalization. The exact mixes contain 378,301/378,341 preference pairs. Chosen responses are slightly *shorter* in aggregate (chosen/rejected mean-word ratio 0.990/0.997), making a general response-length explanation unlikely. At pair level, professional-referral language occurs in 2478 vs 2098 chosen-only/rejected-only pairs at 1B and 2602 vs 1925 at 7B (exact McNemar, BH $q = 4.1 \times 10^{-8} / 2.1 \times 10^{-23}$); per-million-word enrichment remains $1.12\times/1.25\times$.

Model	Stage	Empathy	Validation	Referral	Structuring
1B	base	.217 (.204)	.073 (.073)	.483 (.430)	.757 (.430)
1B	sft	.393 (.393)	.167 (.167)	.697 (.684)	.847 (.767)
1B	dpo	.550 (.510)	.380 (.380)	.933 (.873)	1.000 (.400)
1B	rlvr	.567 (.507)	.410 (.410)	.923 (.836)	1.000 (.440)
1B	instruct	.590 (.550)	.363 (.356)	.900 (.820)	1.000 (.420)
7B	base	.230 (.230)	.087 (.080)	.577 (.524)	.853 (.486)
7B	sft	.493 (.493)	.183 (.183)	.767 (.767)	.787 (.714)
7B	dpo	.597 (.564)	.330 (.330)	.923 (.850)	.977 (.437)
7B	instruct	.617 (.590)	.383 (.370)	.913 (.853)	.970 (.423)

Table 4: Stagewise advice detector-positive rate with the **advice-minus-factual gap in parentheses** ($\tau = 0.82$). The referral gap widens at SFT (+0.25/+0.24, $p < .001$; interaction tests in Table 3); at 1B it widens again at DPO. The structuring gap collapses at DPO because factual answers become listy too. Base checkpoints use a plain QA format (no chat template).

M	Behavior	C-only	R-only	word C/R	BH q
1B	empathy	1768	1634	1.04	.036
1B	validation	183	162	1.12	.282
1B	referral	2478	2098	1.12	$< 10^{-7}$
1B	structuring	27047	21779	1.16	$< 10^{-124}$
7B	empathy	1734	1622	1.05	.074
7B	validation	191	156	1.23	.078
7B	referral	2602	1925	1.25	$< 10^{-22}$
7B	structuring	28466	19838	1.27	$< 10^{-300}$

Table 5: DPO pair-level inference and descriptive length normalization. C/R is the descriptive ratio of phrase occurrences per million whitespace-delimited response words; q is Benjamini–Hochberg adjusted across all eight exact McNemar tests on chosen-only versus rejected-only pairs.

Structuring language is stronger still (27047 vs 21779 and 28466 vs 19838; $q < 10^{-124}$; word-normalized $1.16 \times / 1.27 \times$). Empathy is weakly enriched only at 1B after correction, and validation is not significant (Table 5). These paired co-occurrences are compatible with, but do not attribute or directly validate, the checkpoint changes.

RLVR phrase search: near-zero counts. The RLVR datasets (GSM/MATH/IF) contain essentially none of the advice phrasings: every referral, empathy, and validation phrase counts 0–1 across RLVR–GSM, RLVR–MATH, and the IF-constraints set (only generic markers appear: “first,” 881, “here are some” 48 in the IF set). We find no direct lexical evidence that the searched RLVR sets introduce these behaviors. Their stability through the RLVR stage (Table 4) is consistent with retention rather than new introduction at RLVR.

Pretraining: not a verbatim echo (secondary, hedged). Against the exact Stage-1 corpus, advice answers are more novel than factual ones (0.556 vs 0.365 at 1B; 0.518 vs 0.371 at 7B). This is verbatim-only, length-confounded, and whole-answer rather than behavior-span, so we read it only as: advice answers are not verbatim OLMo-Mix copies. The base checkpoints’ nonzero detector rates are the stronger evidence that the corresponding markers already occur at Base.

4.4 Detector audit and threshold sensitivity

A condition-stratified random audit samples 120 final-model sentences across all five conditions (40 advice, 30 factual, 20 emotional non-advice, 20 neutral advice, 10 domain factual). The sample contains 56 1B and 64 7B sentences from 97 distinct seed-0 responses; it was not stratified on detector predictions, and multiple sentences can come from one response. Two independent blinded Codex passes agree exactly on 117/120 sentences (97.5%); category agreement is 99.2–100%, with Cohen’s $\kappa = .885/.663/1.000/.978$ for empathy/validation/referral/structuring. Claude Opus 4.8 adjudicates the three disagreements. This documents AI consensus under these annotation prompts and models, not human ground truth.

Against that consensus, the detector has **low precision and category-dependent recall** (Table 6; four-category macro P 0.28, R 0.72, F1 0.36; empathy and validation recall are each 0.50). The errors are also condition-dependent: professional referral produces 12 false positives on advice, one on emotional non-advice, two on neutral advice, and none on factual or domain-factual items; validation’s two false positives are both advice items. Thus subtraction does not guarantee cancellation,

category	P	R	F1	tp/fp/fn
empathy expr.	0.22	0.50	0.31	2/7/2
validation	0.33	0.50	0.40	1/2/1
referral	0.06	1.00	0.12	1/15/0
structuring	0.49	0.87	0.63	26/27/4
macro P 0.28 / R 0.72 / F1 0.36; $n=120$				

Table 6: Detector P/R/F1 against the adjudicated 120-sentence AI consensus ($\tau = 0.82$). The two blinded AI passes agreed on 117/120 items; Opus adjudicated three. This is not human validation. Response-cluster bootstrap intervals are released with the artifact. `crisis_referral`: 0 TP / 11 FP, dropped.

and even cross-condition directions remain provisional. `crisis_referral` again fails (0 TP / 11 FP) and is excluded. The detector substantially overpredicts positives in this audit sample, suggesting that absolute output rates are upward-biased; they are not formal upper bounds because recall is below one. The stagewise results should be read as trajectories of a frozen marker detector, not estimates of semantic behavior prevalence.

The low positive prevalence yields wide uncertainty intervals. A 10,000-draw Bayesian bootstrap over the 97 source responses, rather than 120 independent sentences, gives macro P [.16, .42], R [.57, .87], and F1 [.23, .49]; category-level intervals ship with the artifact. The seed-0 sentence sample does not directly validate answer-level emission. The audit itself is AI-only. We therefore also release a blinded two-human workflow with separate files and adjudication, but do not claim it was completed. The threshold sweep is informative rather than uniformly reassuring (Appendix C): structuring’s advice-neutral contrast stays within .07 in all six model-threshold cells and validation exceeds every control in five of six, but empathy reverses at .84 and referral changes magnitude. These are robustness checks on the detector, not a validated taxonomy.

5 Discussion

From “advice template” to trained marker trajectories. The detector-positive decomposition and stagewise gaps nominate a testable account: relevant language is already detected at Base; SFT widens every advice-minus-factual marker gap (six of eight changes survive correction); DPO then acts selectively by widening validation in both sizes and referral at 1B, and by collapsing the structuring gap (spreading organizers to factual answers).

Matched-stage controls further show a robust 7B SFT increase and DPO reversal in empathy specificity, while referral phrasings show strong chosen-only versus rejected-only pair enrichment and descriptive per-word ratios in the same direction; no RLVR-stage gap change is detected. The output trajectory remains provisional because of measurement error. The strongest evidence is instead exact-match and embedding-threshold independent: the DPO chosen/rejected differential and near-zero expressive and referral counts in RLVR do not use e5, although they still rely on the seeded lexicon and establish co-occurrence rather than causation.

Safety implications. For professional-referral phrasing, the bundle’s safety-relevant marker, exact search identifies a finite set of matching SFT assistant turns and measurable DPO chosen-only pair enrichment. That makes the marker auditable and motivates, but does not prove, a fragility hypothesis: mixture or preference-annotation changes could weaken it. The causal leave-cluster-out test our trace nominates (remove the referral cluster, re-SFT, compare against random-cluster removal) is the direct check.

Methodological lessons. (1) Occurrence-weighted provenance ratios are fragile: report distinct phrase types and a leave-one-out. (2) Advice-side rates across stages are ambiguous; report `stage×condition gaps`. (3) Whole-conversation search miscredits user turns; scope to assistant turns. (4) Cluster inference at the prompt, not the sample.

6 Limitations

- **Observational throughout.** Stage localization and data co-occurrence, including the DPO chosen/rejected differential, establish no causation; the nominated leave-cluster-out re-SFT is future work.
- **Detector and threshold.** The complete 120-sentence AI consensus audit finds low precision and category-dependent recall (referral P 0.06; empathy and validation R 0.50) plus condition-dependent errors. The two blinded passes agree strongly and Opus adjudicates all three disagreements, but this is not human validation. The seed-0 sample contains repeated sentences from 97 responses; response-cluster bootstrap intervals remain wide. Substantial overprediction sug-

557
558
559
560
561
562
563

564
565
566
567
568
569

570
571
572

573
574
575
576
577
578
579
580
581
582
583
584

585
586

587

588
589
590
591
592
593
594
595
596
597
598
599
600
601
602
603
604

gests upward-biased absolute rates, while cross-condition gaps and checkpoint trajectories remain provisional. Empathy/referral are threshold-sensitive; structuring and validation are more stable under the threshold sweep but still show false positives. A provenance-safe two-human workflow is released.

- **Coverage gaps.** Stage-2 mid-training (Dolmino-Mix-1124) is unsearched (no public index); matched controls cover Base, SFT, DPO, and final checkpoints but not the intermediate RLVR checkpoint; base checkpoints use a different (plain QA) prompt format.
- **Phrase sparsity.** The SFT co-occurrence rests on 3–6 distinct 3-word types dominated by one list marker; use the leave-one-out figures.
- **Statistical scope.** We apply Benjamini–Hochberg correction across the full 28-test advice/factual adjacent-stage×behavior grid and, separately, across the 24 final-condition and 24 matched-stage tests. Six of eight Base→SFT advice/factual changes survive; no matched-stage validation effect survives at $q < .05$. Five canonical seeds and four extension seeds bound but do not eliminate sampling variance; prompt counts are modest; one model family, two sizes, one decoding configuration; no scaling claim.
- **Novelty** is verbatim-only, length-confounded, and whole-answer.

7 Conclusion

Machine-advice language in OLMo-2 is not one indivisible template. A frozen detector yields distinct affect-, form-, and domain-linked trajectories: all eight advice–factual marker gaps widen at SFT (six survive multiplicity correction), while DPO widens validation in both sizes and referral at 1B, and spreads structuring markers to factual answers. Matched-stage controls show that the most threshold-robust specificity result is a 7B SFT increase followed by a DPO reversal in advice-versus-emotional empathy. The 120-sentence audit, however, finds enough condition-dependent false positives that these output trajectories must remain provisional. The strongest result is embedding-threshold-independent exact matching at the DPO stage: professional-referral and structuring phrases are significantly more common in chosen-only than

rejected-only pairs, with descriptive per-word ratios in the same direction. Searched RLVR data contain almost no referral, empathy, or validation phrases, although generic structuring markers remain. The paper’s contribution is therefore both a trace and an audit of where that trace is trustworthy. We release the tracer, battery, AI consensus labels, Opus decisions, and machine-readable results. The nominated next steps are causal and measurement-side: leave-cluster-out re-SFT against random-cluster controls (cf. [Xiao and Aranguri, 2026](#)), completion of the stratified two-annotator validation, and a self-hosted Stage-2 index.

Ethics Statement

Data and human subjects. This work uses only publicly released artifacts (OLMo-Mix-1124 via the public infinigram API; the OLMo-2 Tulu-3 SFT mixtures, DPO preference mixes, and RLVR sets; open-weight OLMo-2 models and their public intermediate checkpoints). We collected no new data from people. All 140 prompts are short, author-written strings describing hypothetical situations with no personal information. The study involves no human subjects and needed no ethics-board review.

Sensitive domain. Our prompts touch emotional-support, health, and relationship topics, and our analysis concerns safety-relevant moves such as professional referrals. The audit sample shows substantial detector overprediction, suggesting upward-biased absolute rates; we also drop `crisis_referral`. Nothing here should be read as evidence that the studied models reliably provide crisis support; they were not evaluated for clinical accuracy and are not mental-health tools.

Dual use. Making the provenance of a safety behavior inspectable could in principle guide its removal from a training mixture; we frame this as a risk to monitor, show co-occurrence rather than causation, and release an analysis tool, not any model or dataset that could cause direct harm.

Use of AI Assistants

AI coding and writing assistants were used to implement the analysis tooling and to draft and edit the manuscript. Two independent blinded Codex passes annotated the 120-sentence detector audit;

605
606
607
608
609
610
611
612
613
614
615
616
617
618

619

620
621
622
623
624
625
626
627
628
629
630

631
632
633
634
635
636
637
638
639
640

641
642
643
644
645
646

647

648
649
650
651

Claude Opus 4.8 adjudicated their three disagreements. We report this provenance explicitly and do not describe the audit as human validation. The author designed the study, chose the methodology and metrics, and made the scientific decisions, including dropping `crisis_referral` and choosing the statistical unit. All reported numbers were computed by released code from released artifacts and checked against them.

References

Stella Biderman, Hailey Schoelkopf, Quentin Gregory Anthony, Herbie Bradley, Kyle O’Brien, Eric Hallahan, Mohammad Aflah Khan, Shivanshu Purohit, Usven Sai Prashanth, Edward Raff, Aviya Skowron, Lintang Sutawika, and Oskar Van Der Wal. 2023. [Pythia: A suite for analyzing large language models across training and scaling](#). In *Proceedings of the 40th International Conference on Machine Learning*, volume 202 of *Proceedings of Machine Learning Research*, pages 2397–2430. PMLR.

Sreyan Ghosh, Chandra Kiran Reddy Evuru, Sonal Kumar, S. Rameswaran, Deepali Aneja, Zeyu Jin, Ramani Duraiswami, and Dinesh Manocha. 2024. [A closer look at the limitations of instruction tuning](#). In *Proceedings of the 41st International Conference on Machine Learning (ICML)*.

Andreas Köpf, Yannic Kilcher, Dimitri von Rütte, Sotiris Anagnostidis, Zhi-Rui Tam, Keith Stevens, Abdullah Barhoum, Nguyen Minh Duc, Oliver Stanley, Richárd Nagyfi, Shahul ES, Sameer Suri, David Glushkov, Arnav Dantuluri, Andrew Maguire, Christoph Schuhmann, Huu Nguyen, and Alexander Mattick. 2023. [OpenAssistant conversations – democratizing large language model alignment](#). In *Advances in Neural Information Processing Systems 36 (NeurIPS), Datasets and Benchmarks Track*.

Nathan Lambert, Jacob Morrison, Valentina Pyatkin, Shengyi Huang, Hamish Ivison, Faeze Brahman, Lester James V. Miranda, Alisa Liu, Nouha Dziri, Shane Lyu, Yuling Gu, Saumya Malik, Victoria Graf, Jena D. Hwang, Jiangjiang Yang, Ronan Le Bras, Oyvind Tafjord, Chris Wilhelm, Luca Soldaini, and 4 others. 2024. [Tulu 3: Pushing frontiers in open language model post-training](#). *Preprint*, arXiv:2411.15124.

Jiacheng Liu, Taylor Blanton, Yanai Elazar, Sewon Min, Yen-Sung Chen, Arnavi Chheda-Kothary, Huy Tran, Byron Bischoff, Eric Marsh, Michael Schmitz, Cassidy Trier, Aaron Sarnat, Jenna James, Jon Borchardt, Bailey Kuehl, Evie Yu-Yen Cheng, Karen Farley, Taira Anderson, David Albright, and 11 others. 2025. [OLMoTrace: Tracing language model outputs back to trillions of training tokens](#). In *Proceedings of the 63rd Annual Meeting of the Association for Computational Linguistics (ACL): System Demonstrations*,

pages 178–188, Vienna, Austria. Association for Computational Linguistics.

Jiacheng Liu, Sewon Min, Luke Zettlemoyer, Yejin Choi, and Hannaneh Hajishirzi. 2024. [Infini-gram: Scaling unbounded \$n\$ -gram language models to a trillion tokens](#). In *First Conference on Language Modeling (COLM)*.

Marco Tulio Ribeiro, Tongshuang Wu, Carlos Guestrin, and Sameer Singh. 2020. [Beyond accuracy: Behavioral testing of NLP models with CheckList](#). In *Proceedings of the 58th Annual Meeting of the Association for Computational Linguistics*, pages 4902–4912. Association for Computational Linguistics.

Team OLMo, Pete Walsh, Luca Soldaini, Dirk Groeneveld, Kyle Lo, Shane Arora, Akshita Bhagia, Yuling Gu, Shengyi Huang, Matt Jordan, Nathan Lambert, Dustin Schwenk, Oyvind Tafjord, Taira Anderson, David Atkinson, Faeze Brahman, Christopher Clark, Pradeep Dasigi, Nouha Dziri, and 24 others. 2025. [2 OLMo 2 Furious](#). *Preprint*, arXiv:2501.00656. A shorter version appeared at COLM 2025.

Liang Wang, Nan Yang, Xiaolong Huang, Linjun Yang, Rangan Majumder, and Furu Wei. 2024. [Multilingual e5 text embeddings: A technical report](#). *Preprint*, arXiv:2402.05672.

Mengzhou Xia, Mikel Artetxe, Chunting Zhou, Xi Victoria Lin, Ramakanth Pasunuru, Danqi Chen, Luke Zettlemoyer, and Veselin Stoyanov. 2023. [Training trajectories of language models across scales](#). In *Proceedings of the 61st Annual Meeting of the Association for Computational Linguistics (Volume 1: Long Papers)*, pages 13711–13738. Association for Computational Linguistics.

Mengzhou Xia, Sadhika Malladi, Suchin Gururangan, Sanjeev Arora, and Danqi Chen. 2024. [LESS: Selecting influential data for targeted instruction tuning](#). In *Proceedings of the 41st International Conference on Machine Learning*.

Frank Xiao and Santiago Aranguri. 2026. [Probe-based data attribution: Discovering and mitigating undesirable behaviors in LLM post-training](#). *Preprint*, arXiv:2602.11079. ArXiv preprint, v3.

Chunting Zhou, Pengfei Liu, Puxin Xu, Srinu Iyer, Jiao Sun, Yuning Mao, Xuezhe Ma, Avia Efrat, Ping Yu, Lili Yu, Susan Zhang, Gargi Ghosh, Mike Lewis, Luke Zettlemoyer, and Omer Levy. 2023. [LIMA: Less is more for alignment](#). In *Advances in Neural Information Processing Systems 36 (NeurIPS)*.

A Prompts and decoding configuration

All answers use nucleus sampling (temperature 0.7, top- p 0.9, `max_new_tokens=256`, `do_sample=True`), the chat template where available (plain `Question: /Answer:` for base checkpoints), and five seeds per prompt in the

canonical grid. The battery: 60 advice, 30 factual, 20 emotional non-advice, 20 neutral advice, 10 domain factual; the complete lists ship with the artifact. The Base/SFT/DPO matched-control extension uses four seeds; all balanced matched-stage tests use seeds 0–3 for every condition.

Examples.

- **Advice:** I’ve been feeling really anxious lately and can’t sleep. Any advice?
- **Factual:** How does photosynthesis work?
- **Emotional non-advice:** Describe the common symptoms of loneliness.
- **Neutral advice:** How should I organize my bookshelf?
- **Domain factual:** What are common clinical treatments for insomnia?

B Behavior lexicon and detector threshold

A category fires for an answer if *either* (i) any lexicon phrase occurs as a case-insensitive substring, *or* (ii) any sentence has cosine similarity $\geq \tau$ to a seed exemplar (intfloat/multilingual-e5-small, query:passage: prefixes), $\tau = 0.82$ primary.

Lexicon (substring match).

- **empathy expression** (empathy_opener):
i’m sorry to hear, i am sorry to hear, i’m really sorry, that sounds really, that sounds very, i can understand, i can imagine, it’s understandable, that must be, i’m here for you, i hear you, it sounds like you
- **validation:** your feelings are valid, it’s okay to feel, it’s normal to feel, it’s completely normal, what you’re feeling, there’s nothing wrong with, it’s natural to
- **professional referral** (disclaimer):
consult a, talk to a, speak with a, see a doctor, seek medical,

i am not a doctor, i’m not a doctor, medical professional, healthcare professional, mental health professional, seek professional, professional help

- **crisis_referral** (dropped after 0 TP / 11 FP in the 120-sentence audit): crisis, hotline, helpline, 988, call 911, emergency services, suicide
- **structuring language** (structure):
here are some, here are a few, consider the following, steps you can take, a few things you can, first,, secondly,

e5 seed exemplars. Empathy: “I’m so sorry you’re going through this”; “That sounds incredibly hard.” Validation: “Your feelings are completely valid”; “It’s totally normal to feel this way.” Referral: “You should consult a qualified professional”; “Please talk to your doctor about this.” Structuring: “Here are a few steps you can take”; “Consider the following suggestions.”

C Threshold sweep

Table 7 reports the exact contrasts underlying the decomposition at $\tau = 0.80/0.82/0.84$. Structuring is stable: advice and neutral-advice remain within .05 in every cell. Validation exceeds its strongest control in all six cells. Empathy’s emotional-minus-advice contrast reverses at .84, while referral’s advice-minus-domain contrast changes magnitude. The primary $\tau = .82$ was frozen before the 140-prompt battery, but its qualitative decomposition should not be generalized across detector thresholds.

M	Contrast	.80	.82	.84
1B	empathy: emot.—adv.	.030	.200	-.107
7B		.030	.143	-.280
1B	validation: adv.—max ctrl.	.030	.243	.107
7B		.047	.253	.170
1B	referral: adv.—domain	-.003	.080	.093
7B		.017	.093	.280
1B	structuring: adv.—neutral	.000	.000	.047
7B		.000	-.030	-.010

Table 7: Threshold robustness of the four decomposition contrasts. Positive values support the primary-threshold reading except for structuring, where values near zero indicate the same rate on advice and neutral advice.

D Matched-control checkpoint interactions

Table 8 reports every planned interaction in the four-seed Base/SFT/DPO extension. Columns Base, SFT, and DPO are advice-minus-control gaps; transition columns are changes in those gaps with BH-adjusted q in parentheses. The family contains all 24 displayed transition tests.

E Equivalent-count sensitivity analysis

For readers who want familiar binomial statistics, this table binarizes each prompt’s five-generation mean into an equivalent count ($k = \text{round}(\bar{p} \cdot n)$) and applies Wilson intervals, Fisher exact tests, and Haldane–Anscombe odds ratios. **Caveat: these equivalent counts are not independent Bernoulli trials** (each aggregates five correlated generations), so this is a sensitivity analysis only; the prompt-clustered permutation tests in the main text are the primary inference.

F Per-phrase recoverability and leave-one-out

Per-phrase assistant-only counts against each model’s exact mixture (–0225, 866,138 docs, 1B; `tulu-3-sft-olmo-2-mixture`, 939,344, 7B); `infl = blob/assistant`.

Length-matched buckets (occurrence-weighted). Behavioral vs topical per-million for 1B: 2-word 1884.3 vs 1760.2 ($1.1\times$, $n_{\text{occ}}=14$), 3-word 12900.2 vs 629.8 ($20.5\times$, 45), 4-word 1596.9 vs 11.7 ($136.9\times$, 30). For 7B, 2-word 1784.5 vs 996.8 ($1.8\times$, 20), 3-word 9354.4 vs 1280.4 ($7.3\times$, 55), 4-word 1396.6 vs 96.2 ($14.5\times$, 24). **Distinct-phrase leave-one-out (3-word):** dropping “here are some” moves the mean from 5455.2 to 1305.2/M at 1B (3 types) and from 3026.0 to 1065.9/M at 7B (6 types).

G Response-cluster audit uncertainty

The 120 audit sentences come from 97 seed-0 responses. To preserve dependence among sentences from the same response, we assign $\text{Dirichlet}(1)$ weights to response keys in a 10,000-draw Bayesian bootstrap and recompute each metric. Table 11 reports percentile intervals. These intervals quantify sampling uncertainty around the AI consensus labels; they do not turn the audit into human validation.

H Statistical formulas

Primary inference uses two-sided prompt-clustered permutation tests on prompt-level means (10,000 draws), with Monte Carlo $p = (b + 1)/(B + 1)$; $p < 10^{-4}$ denotes zero raw exceedances. Appendix E reports Wilson 95% intervals, two-sided Fisher tests, and Haldane–Anscombe odds ratios on equivalent counts. The distinct-phrase leave-one-out removes the most frequent 3-word phrase before recomputing the mean behavioral per-million. DPO comparisons use exact McNemar tests on chosen-only versus rejected-only pairs and phrase occurrences per million whitespace-delimited response words. Released scripts derive all statistics from the artifact.

M	Marker	Control	Base	SFT	DPO	Base→SFT	SFT→DPO
1B	referral	domain factual	.188	.308	.167	+.121 (.414)	-.142 (.296)
1B	empathy	emotional	.017	.225	-.242	+.208 (.095)	-.467 (.001)
1B	structuring	neutral advice	-.013	-.050	.000	-.037 (.653)	+.050 (.475)
1B	validation	domain factual	.071	.146	.375	+.075 (.475)	+.229 (.095)
1B	validation	emotional	.021	.121	.225	+.100 (.293)	+.104 (.296)
1B	validation	neutral advice	.058	.171	.350	+.113 (.189)	+.179 (.059)
7B	referral	domain factual	.321	.067	.200	-.254 (.166)	+.133 (.293)
7B	empathy	emotional	.108	.467	-.175	+.358 (.016)	-.642 (.001)
7B	structuring	neutral advice	.050	-.158	-.025	-.208 (.026)	+.133 (.095)
7B	validation	domain factual	.092	.175	.317	+.083 (.475)	+.142 (.296)
7B	validation	emotional	.054	.125	.179	+.071 (.414)	+.054 (.537)
7B	validation	neutral advice	.092	.163	.292	+.071 (.414)	+.129 (.189)

Table 8: Complete matched-control stage interactions at $\tau = .82$ (10k prompt-clustered permutations). Four of 24 transition tests have $q < .05$. Only the 7B empathy Base→SFT and SFT→DPO effects retain $q < .05$ at all three detector thresholds.

M	Behavior	Advice	Factual	OR	Fisher p
1B	empathy	35/60	1/30	27.4	1.4×10^{-7}
1B	validation	22/60	0/30	35.6	4.2×10^{-5}
1B	referral	54/60	2/30	95.6	3.1×10^{-15}
1B	structuring	60/60	17/30	93.3	7.3×10^{-8}
7B	empathy	37/60	1/30	31.4	2.2×10^{-8}
7B	validation	23/60	0/30	38.2	1.9×10^{-5}
7B	referral	55/60	2/30	115.0	5.7×10^{-16}
7B	structuring	58/60	16/30	20.6	1.2×10^{-6}

Table 9: Equivalent-count sensitivity analysis (advice vs factual, $\tau = 0.82$). Directionally identical to the permutation results; not valid as primary inference (correlated generations within prompts).

Model	Phrase	Asst.	Blob	Infl.
1B	first,	44422	52676	1.19
1B	here are some	11914	13276	1.11
1B	here are a few	5421	6056	1.12
1B	healthcare professional	3075	3216	1.05
1B	mental health professional	2109	2186	1.04
1B	professional help	1765	1825	1.03
1B	seek professional	1353	1377	1.02
1B	it sounds like you	1130	1164	1.03
1B	consult a	891	978	1.10
1B	it's understandable	218	227	1.04
1B	it's natural to	152	157	1.03
7B	here are some	12049	13432	1.11
7B	here are a few	5511	6156	1.12
7B	healthcare professional	3106	3309	1.07
7B	i'm really sorry	2369	2378	1.00
7B	mental health professional	2120	2214	1.04
7B	professional help	1774	1838	1.04
7B	seek professional	1355	1379	1.02
7B	it sounds like you	1139	1184	1.04
7B	consult a	927	1031	1.11
7B	talk to a	286	408	1.43
7B	it's completely normal	78	103	1.32

Table 10: Per-phrase SFT-assistant recoverability (representative rows). The 3 (1B) / 6 (7B) distinct 3-word types drive Section 4.3; “here are some” dominates the 3-word mean.

Category	Precision 95%	Recall 95%	F1 95%
empathy expression	[.04,.46]	[.10,.90]	[.05,.57]
validation	[.02,.84]	[.03,.98]	[.02,.83]
referral	[.00,.22]	[1.00,1.00]	[.00,.36]
structuring	[.35,.63]	[.74,.96]	[.49,.74]
macro	[.16,.42]	[.57,.87]	[.23,.49]

Table 11: Bayesian response-cluster bootstrap intervals for the 120-sentence AI consensus audit. Referral recall is degenerate because the single positive gold item is detected in every draw; its precision and F1 remain highly uncertain.